Mode 11 — Square-BFP (centered / opt)

C-DP16 (C8×C8) → DP32 (8×8) → DP16 (8×8) → DP16 (8×4) → DP8 (8×4) → block exponents → DP8 (4×4) → centered add-and-square

Step 1 — C-DP16 (C8×C8)

$$\text{C-DP16}\,(\text{C8}\times\text{C8}) = D = \sum_{i=0}^{15} a_i\,b_i, \qquad a_i = a_i^{re} + \mathrm{j}\,a_i^{im}, \quad b_i = b_i^{re} + \mathrm{j}\,b_i^{im}$$

Step 2 — DP32 (8×8)

$$\mathrm{Re}(D) = \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{re} - \sum_{i=0}^{15} a_i^{im} b_i^{im}}_{\text{DP32}\,(8\times8)}$$

$$\mathrm{Im}(D) = \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{im} + \sum_{i=0}^{15} a_i^{im} b_i^{re}}_{\text{DP32}\,(8\times8)}$$

Step 3 — DP16 (8×8)

$$\mathrm{Re}(D) = \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{re}}_{\text{DP16}\,(8\times8)} - \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{im}}_{\text{DP16}\,(8\times8)}$$

$$\mathrm{Im}(D) = \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{im}}_{\text{DP16}\,(8\times8)} + \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{re}}_{\text{DP16}\,(8\times8)}$$

Step 4 — DP16 (8×4)

$$\mathrm{Re}(D) = 2^4 \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{re,hi}}_{\text{DP16}\,(8\times4)} + \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{re,lo}}_{\text{DP16}\,(8\times4)} - 2^4 \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{im,hi}}_{\text{DP16}\,(8\times4)} - \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{im,lo}}_{\text{DP16}\,(8\times4)}$$

$$\mathrm{Im}(D) = 2^4 \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{im,hi}}_{\text{DP16}\,(8\times4)} + \underbrace{\sum_{i=0}^{15} a_i^{re} b_i^{im,lo}}_{\text{DP16}\,(8\times4)} + 2^4 \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{re,hi}}_{\text{DP16}\,(8\times4)} + \underbrace{\sum_{i=0}^{15} a_i^{im} b_i^{re,lo}}_{\text{DP16}\,(8\times4)}$$

Step 5 — DP8 (8×4)

$$\begin{aligned}\mathrm{Re}(D) = {} & 2^4 \underbrace{\sum_{i=0}^{7} a_i^{re} b_i^{re,hi}}_{\text{DP8}\,(8\times4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{re} b_i^{re,hi}}_{\text{DP8}\,(8\times4)} + \underbrace{\sum_{i=0}^{7} a_i^{re} b_i^{re,lo}}_{\text{DP8}\,(8\times4)} + \underbrace{\sum_{i=8}^{15} a_i^{re} b_i^{re,lo}}_{\text{DP8}\,(8\times4)} \\ & - 2^4 \underbrace{\sum_{i=0}^{7} a_i^{im} b_i^{im,hi}}_{\text{DP8}\,(8\times4)} - 2^4 \underbrace{\sum_{i=8}^{15} a_i^{im} b_i^{im,hi}}_{\text{DP8}\,(8\times4)} - \underbrace{\sum_{i=0}^{7} a_i^{im} b_i^{im,lo}}_{\text{DP8}\,(8\times4)} - \underbrace{\sum_{i=8}^{15} a_i^{im} b_i^{im,lo}}_{\text{DP8}\,(8\times4)}\end{aligned}$$

$$\mathrm{Im}(D) = 2^4 \underbrace{\sum_{i=0}^{7} a_i^{re} b_i^{im,hi}}_{\text{DP8}\,(8\times4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_i^{re} b_i^{im,hi}}_{\text{DP8}\,(8\times4)} + \underbrace{\sum_{i=0}^{7} a_i^{re} b_i^{im,lo}}_{\text{DP8}\,(8\times4)} + \underbrace{\sum_{i=8}^{15} a_i^{re} b_i^{im,lo}}_{\text{DP8}\,(8\times4)}$$

$$+\,2^4\underbrace{\sum_{i=0}^{7} a_i^{im} b_i^{re,hi}}_{\text{DP8 }(8\times4)} + 2^4\underbrace{\sum_{i=8}^{15} a_i^{im} b_i^{re,hi}}_{\text{DP8 }(8\times4)} + \underbrace{\sum_{i=0}^{7} a_i^{im} b_i^{re,lo}}_{\text{DP8 }(8\times4)} + \underbrace{\sum_{i=8}^{15} a_i^{im} b_i^{re,lo}}_{\text{DP8 }(8\times4)}$$

Step 6 — Block exponents (one exponent block per DP8 (8×4); Re/Im share E_k)

$$M^{re} = 2^{-\delta_0}\Big(2^4\sum_{i=0}^{7} a_i^{re} b_i^{re,hi} + \sum_{i=0}^{7} a_i^{re} b_i^{re,lo} - 2^4\sum_{i=0}^{7} a_i^{im} b_i^{im,hi} - \sum_{i=0}^{7} a_i^{im} b_i^{im,lo}\Big) + 2^{-\delta_1}\Big(\cdots\ i\in[8,15]\ \cdots\Big)$$

$$M^{im} = 2^{-\delta_0}\Big(2^4\sum_{i=0}^{7} a_i^{re} b_i^{im,hi} + \sum_{i=0}^{7} a_i^{re} b_i^{im,lo} + 2^4\sum_{i=0}^{7} a_i^{im} b_i^{re,hi} + \sum_{i=0}^{7} a_i^{im} b_i^{re,lo}\Big) + 2^{-\delta_1}\Big(\cdots\ i\in[8,15]\ \cdots\Big)$$

$E_k = E_{a,k} + E_{b,k}$; $E = \max(E_0, E_1)$, $\delta_k = E - E_k$; truncating shifts, on block sums only; $\mathrm{Re}(\hat{D}) = M^{re}\cdot 2^E$, $\mathrm{Im}(\hat{D}) = M^{im}\cdot 2^E$.

Step 7 — DP8 (4×4)

$$\begin{aligned}
M^{re} = {} & 2^{-\delta_0}\,2^8\underbrace{\sum_{i=0}^{7} a_{H,i}^{re}\, b_i^{re,hi}}_{\text{DP8 }(4\times4)} + 2^{-\delta_0}\,2^4\underbrace{\sum_{i=0}^{7} a_{L,i}^{re}\, b_i^{re,hi}}_{\text{DP8 }(4\times4)} + 2^{-\delta_1}\,2^8\underbrace{\sum_{i=8}^{15} a_{H,i}^{re}\, b_i^{re,hi}}_{\text{DP8 }(4\times4)} + 2^{-\delta_1}\,2^4\underbrace{\sum_{i=8}^{15} a_{L,i}^{re}\, b_i^{re,hi}}_{\text{DP8 }(4\times4)} \\
& + 2^{-\delta_0}\,2^4\underbrace{\sum_{i=0}^{7} a_{H,i}^{re}\, b_i^{re,lo}}_{\text{DP8 }(4\times4)} + 2^{-\delta_0}\underbrace{\sum_{i=0}^{7} a_{L,i}^{re}\, b_i^{re,lo}}_{\text{DP8 }(4\times4)} + 2^{-\delta_1}\,2^4\underbrace{\sum_{i=8}^{15} a_{H,i}^{re}\, b_i^{re,lo}}_{\text{DP8 }(4\times4)} + 2^{-\delta_1}\underbrace{\sum_{i=8}^{15} a_{L,i}^{re}\, b_i^{re,lo}}_{\text{DP8 }(4\times4)} \\
& - 2^{-\delta_0}\,2^8\underbrace{\sum_{i=0}^{7} a_{H,i}^{im}\, b_i^{im,hi}}_{\text{DP8 }(4\times4)} - 2^{-\delta_0}\,2^4\underbrace{\sum_{i=0}^{7} a_{L,i}^{im}\, b_i^{im,hi}}_{\text{DP8 }(4\times4)} - 2^{-\delta_1}\,2^8\underbrace{\sum_{i=8}^{15} a_{H,i}^{im}\, b_i^{im,hi}}_{\text{DP8 }(4\times4)} - 2^{-\delta_1}\,2^4\underbrace{\sum_{i=8}^{15} a_{L,i}^{im}\, b_i^{im,hi}}_{\text{DP8 }(4\times4)} \\
& - 2^{-\delta_0}\,2^4\underbrace{\sum_{i=0}^{7} a_{H,i}^{im}\, b_i^{im,lo}}_{\text{DP8 }(4\times4)} - 2^{-\delta_0}\underbrace{\sum_{i=0}^{7} a_{L,i}^{im}\, b_i^{im,lo}}_{\text{DP8 }(4\times4)} - 2^{-\delta_1}\,2^4\underbrace{\sum_{i=8}^{15} a_{H,i}^{im}\, b_i^{im,lo}}_{\text{DP8 }(4\times4)} - 2^{-\delta_1}\underbrace{\sum_{i=8}^{15} a_{L,i}^{im}\, b_i^{im,lo}}_{\text{DP8 }(4\times4)}
\end{aligned}$$

$$\begin{aligned}
M^{im} = {} & 2^{-\delta_0}\,2^8\underbrace{\sum_{i=0}^{7} a_{H,i}^{re}\, b_i^{im,hi}}_{\text{DP8 }(4\times4)} + 2^{-\delta_0}\,2^4\underbrace{\sum_{i=0}^{7} a_{L,i}^{re}\, b_i^{im,hi}}_{\text{DP8 }(4\times4)} + 2^{-\delta_1}\,2^8\underbrace{\sum_{i=8}^{15} a_{H,i}^{re}\, b_i^{im,hi}}_{\text{DP8 }(4\times4)} + 2^{-\delta_1}\,2^4\underbrace{\sum_{i=8}^{15} a_{L,i}^{re}\, b_i^{im,hi}}_{\text{DP8 }(4\times4)} \\
& + 2^{-\delta_0}\,2^4\underbrace{\sum_{i=0}^{7} a_{H,i}^{re}\, b_i^{im,lo}}_{\text{DP8 }(4\times4)} + 2^{-\delta_0}\underbrace{\sum_{i=0}^{7} a_{L,i}^{re}\, b_i^{im,lo}}_{\text{DP8 }(4\times4)} + 2^{-\delta_1}\,2^4\underbrace{\sum_{i=8}^{15} a_{H,i}^{re}\, b_i^{im,lo}}_{\text{DP8 }(4\times4)} + 2^{-\delta_1}\underbrace{\sum_{i=8}^{15} a_{L,i}^{re}\, b_i^{im,lo}}_{\text{DP8 }(4\times4)} \\
& + 2^{-\delta_0}\,2^8\underbrace{\sum_{i=0}^{7} a_{H,i}^{im}\, b_i^{re,hi}}_{\text{DP8 }(4\times4)} + 2^{-\delta_0}\,2^4\underbrace{\sum_{i=0}^{7} a_{L,i}^{im}\, b_i^{re,hi}}_{\text{DP8 }(4\times4)} + 2^{-\delta_1}\,2^8\underbrace{\sum_{i=8}^{15} a_{H,i}^{im}\, b_i^{re,hi}}_{\text{DP8 }(4\times4)} + 2^{-\delta_1}\,2^4\underbrace{\sum_{i=8}^{15} a_{L,i}^{im}\, b_i^{re,hi}}_{\text{DP8 }(4\times4)} \\
& + 2^{-\delta_0}\,2^4\underbrace{\sum_{i=0}^{7} a_{H,i}^{im}\, b_i^{re,lo}}_{\text{DP8 }(4\times4)} + 2^{-\delta_0}\underbrace{\sum_{i=0}^{7} a_{L,i}^{im}\, b_i^{re,lo}}_{\text{DP8 }(4\times4)} + 2^{-\delta_1}\,2^4\underbrace{\sum_{i=8}^{15} a_{H,i}^{im}\, b_i^{re,lo}}_{\text{DP8 }(4\times4)} + 2^{-\delta_1}\underbrace{\sum_{i=8}^{15} a_{L,i}^{im}\, b_i^{re,lo}}_{\text{DP8 }(4\times4)}
\end{aligned}$$

Step 8 — Centered add-and-square (subtract 8 from unsigned nibbles a_L, b^{lo}; per DP8 (4×4) add $c = 0/512/2048$ for 0/1/2 unsigned nibbles; the $-\sum a^{im}b^{im}$ group keeps add-and-square but the whole block carries a minus)

$$M^{re} = 2^{-\delta_0}\,2^8\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a_{H,i}^{re}+b_i^{re,hi})^2 - \sum_{i=0}^{7}(a_{H,i}^{re})^2 - \sum_{i=0}^{7}(b_i^{re,hi})^2\Big)}_{\text{DP8 }(4\times4)}$$

$$+\ 2^{-\delta_0}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{re}_{L,i}-8)+b^{re,hi}_i\big)^2-\sum_{i=0}^{7}(a^{re}_{L,i}-8)^2-\sum_{i=0}^{7}(b^{re,hi}_i-8)^2+512\Big)}_{\text{DP8 (4×4)}}$$

$$+\ 2^{-\delta_1}\, 2^8\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}(a^{re}_{H,i}+b^{re,hi}_i)^2-\sum_{i=8}^{15}(a^{re}_{H,i})^2-\sum_{i=8}^{15}(b^{re,hi}_i)^2\Big)}_{\text{DP8 (4×4)}}$$

$$+\ 2^{-\delta_1}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}\big((a^{re}_{L,i}-8)+b^{re,hi}_i\big)^2-\sum_{i=8}^{15}(a^{re}_{L,i}-8)^2-\sum_{i=8}^{15}(b^{re,hi}_i-8)^2+512\Big)}_{\text{DP8 (4×4)}}$$

$$+\ 2^{-\delta_0}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big(a^{re}_{H,i}+(b^{re,lo}_i-8)\big)^2-\sum_{i=0}^{7}(a^{re}_{H,i}-8)^2-\sum_{i=0}^{7}(b^{re,lo}_i-8)^2+512\Big)}_{\text{DP8 (4×4)}}$$

$$+\ 2^{-\delta_0}\ \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{re}_{L,i}-8)+(b^{re,lo}_i-8)\big)^2-\sum_{i=0}^{7}(a^{re}_{L,i}-16)^2-\sum_{i=0}^{7}(b^{re,lo}_i-16)^2+2048\Big)}_{\text{DP8 (4×4)}}$$

$$+\ 2^{-\delta_1}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}\big(a^{re}_{H,i}+(b^{re,lo}_i-8)\big)^2-\sum_{i=8}^{15}(a^{re}_{H,i}-8)^2-\sum_{i=8}^{15}(b^{re,lo}_i-8)^2+512\Big)}_{\text{DP8 (4×4)}}$$

$$+\ 2^{-\delta_1}\ \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}\big((a^{re}_{L,i}-8)+(b^{re,lo}_i-8)\big)^2-\sum_{i=8}^{15}(a^{re}_{L,i}-16)^2-\sum_{i=8}^{15}(b^{re,lo}_i-16)^2+2048\Big)}_{\text{DP8 (4×4)}}$$

$$-\ 2^{-\delta_0}\, 2^8\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}(a^{im}_{H,i}+b^{im,hi}_i)^2-\sum_{i=0}^{7}(a^{im}_{H,i})^2-\sum_{i=0}^{7}(b^{im,hi}_i)^2\Big)}_{\text{DP8 (4×4)}}$$

$$-\ 2^{-\delta_0}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{im}_{L,i}-8)+b^{im,hi}_i\big)^2-\sum_{i=0}^{7}(a^{im}_{L,i}-8)^2-\sum_{i=0}^{7}(b^{im,hi}_i-8)^2+512\Big)}_{\text{DP8 (4×4)}}$$

$$-\ 2^{-\delta_1}\, 2^8\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}(a^{im}_{H,i}+b^{im,hi}_i)^2-\sum_{i=8}^{15}(a^{im}_{H,i})^2-\sum_{i=8}^{15}(b^{im,hi}_i)^2\Big)}_{\text{DP8 (4×4)}}$$

$$-\ 2^{-\delta_1}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}\big((a^{im}_{L,i}-8)+b^{im,hi}_i\big)^2-\sum_{i=8}^{15}(a^{im}_{L,i}-8)^2-\sum_{i=8}^{15}(b^{im,hi}_i-8)^2+512\Big)}_{\text{DP8 (4×4)}}$$

$$-\ 2^{-\delta_0}\, 2^4\, \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big(a^{im}_{H,i}+(b^{im,lo}_i-8)\big)^2-\sum_{i=0}^{7}(a^{im}_{H,i}-8)^2-\sum_{i=0}^{7}(b^{im,lo}_i-8)^2+512\Big)}_{\text{DP8 (4×4)}}$$

$$-\ 2^{-\delta_0}\ \underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{im}_{L,i}-8)+(b^{im,lo}_i-8)\big)^2-\sum_{i=0}^{7}(a^{im}_{L,i}-16)^2-\sum_{i=0}^{7}(b^{im,lo}_i-16)^2+2048\Big)}_{\text{DP8 (4×4)}}$$

$$- \; 2^{-\delta_1}\, 2^4\, \tfrac{1}{2}\underbrace{\Big(\sum_{i=8}^{15}\big(a^{im}_{H,i}+(b^{im,lo}_i-8)\big)^2 - \sum_{i=8}^{15}(a^{im}_{H,i}-8)^2 - \sum_{i=8}^{15}(b^{im,lo}_i-8)^2 + 512\Big)}_{\text{DP8}\,(4{\times}4)}$$

$$- \; 2^{-\delta_1}\, \tfrac{1}{2}\underbrace{\Big(\sum_{i=8}^{15}\big((a^{im}_{L,i}-8)+(b^{im,lo}_i-8)\big)^2 - \sum_{i=8}^{15}(a^{im}_{L,i}-16)^2 - \sum_{i=8}^{15}(b^{im,lo}_i-16)^2 + 2048\Big)}_{\text{DP8}\,(4{\times}4)}$$

$$M^{im} = 2^{-\delta_0}\, 2^8\, \tfrac{1}{2}\underbrace{\Big(\sum_{i=0}^{7}(a^{re}_{H,i}+b^{im,hi}_i)^2 - \sum_{i=0}^{7}(a^{re}_{H,i})^2 - \sum_{i=0}^{7}(b^{im,hi}_i)^2\Big)}_{\text{DP8}\,(4{\times}4)}$$

$$+ \; 2^{-\delta_0}\, 2^4\, \tfrac{1}{2}\underbrace{\Big(\sum_{i=0}^{7}\big((a^{re}_{L,i}-8)+b^{im,hi}_i\big)^2 - \sum_{i=0}^{7}(a^{re}_{L,i}-8)^2 - \sum_{i=0}^{7}(b^{im,hi}_i-8)^2 + 512\Big)}_{\text{DP8}\,(4{\times}4)}$$

$$+ \; 2^{-\delta_1}\, 2^8\, \tfrac{1}{2}\underbrace{\Big(\sum_{i=8}^{15}(a^{re}_{H,i}+b^{im,hi}_i)^2 - \sum_{i=8}^{15}(a^{re}_{H,i})^2 - \sum_{i=8}^{15}(b^{im,hi}_i)^2\Big)}_{\text{DP8}\,(4{\times}4)}$$

$$+ \; 2^{-\delta_1}\, 2^4\, \tfrac{1}{2}\underbrace{\Big(\sum_{i=8}^{15}\big((a^{re}_{L,i}-8)+b^{im,hi}_i\big)^2 - \sum_{i=8}^{15}(a^{re}_{L,i}-8)^2 - \sum_{i=8}^{15}(b^{im,hi}_i-8)^2 + 512\Big)}_{\text{DP8}\,(4{\times}4)}$$

$$+ \; 2^{-\delta_0}\, 2^4\, \tfrac{1}{2}\underbrace{\Big(\sum_{i=0}^{7}\big(a^{re}_{H,i}+(b^{im,lo}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{re}_{H,i}-8)^2 - \sum_{i=0}^{7}(b^{im,lo}_i-8)^2 + 512\Big)}_{\text{DP8}\,(4{\times}4)}$$

$$+ \; 2^{-\delta_0}\, \tfrac{1}{2}\underbrace{\Big(\sum_{i=0}^{7}\big((a^{re}_{L,i}-8)+(b^{im,lo}_i-8)\big)^2 - \sum_{i=0}^{7}(a^{re}_{L,i}-16)^2 - \sum_{i=0}^{7}(b^{im,lo}_i-16)^2 + 2048\Big)}_{\text{DP8}\,(4{\times}4)}$$

$$+ \; 2^{-\delta_1}\, 2^4\, \tfrac{1}{2}\underbrace{\Big(\sum_{i=8}^{15}\big(a^{re}_{H,i}+(b^{im,lo}_i-8)\big)^2 - \sum_{i=8}^{15}(a^{re}_{H,i}-8)^2 - \sum_{i=8}^{15}(b^{im,lo}_i-8)^2 + 512\Big)}_{\text{DP8}\,(4{\times}4)}$$

$$+ \; 2^{-\delta_1}\, \tfrac{1}{2}\underbrace{\Big(\sum_{i=8}^{15}\big((a^{re}_{L,i}-8)+(b^{im,lo}_i-8)\big)^2 - \sum_{i=8}^{15}(a^{re}_{L,i}-16)^2 - \sum_{i=8}^{15}(b^{im,lo}_i-16)^2 + 2048\Big)}_{\text{DP8}\,(4{\times}4)}$$

$$+ \; 2^{-\delta_0}\, 2^8\, \tfrac{1}{2}\underbrace{\Big(\sum_{i=0}^{7}(a^{im}_{H,i}+b^{re,hi}_i)^2 - \sum_{i=0}^{7}(a^{im}_{H,i})^2 - \sum_{i=0}^{7}(b^{re,hi}_i)^2\Big)}_{\text{DP8}\,(4{\times}4)}$$

$$+ \; 2^{-\delta_0}\, 2^4\, \tfrac{1}{2}\underbrace{\Big(\sum_{i=0}^{7}\big((a^{im}_{L,i}-8)+b^{re,hi}_i\big)^2 - \sum_{i=0}^{7}(a^{im}_{L,i}-8)^2 - \sum_{i=0}^{7}(b^{re,hi}_i-8)^2 + 512\Big)}_{\text{DP8}\,(4{\times}4)}$$

$$+ \; 2^{-\delta_1}\, 2^8\, \tfrac{1}{2}\underbrace{\Big(\sum_{i=8}^{15}(a^{im}_{H,i}+b^{re,hi}_i)^2 - \sum_{i=8}^{15}(a^{im}_{H,i})^2 - \sum_{i=8}^{15}(b^{re,hi}_i)^2\Big)}_{\text{DP8}\,(4{\times}4)}$$

$$+\ 2^{-\delta_1}\,2^4\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}\big((a^{im}_{L,i}-8)+b^{re,hi}_i\big)^2-\sum_{i=8}^{15}(a^{im}_{L,i}-8)^2-\sum_{i=8}^{15}(b^{re,hi}_i-8)^2+512\Big)}_{\text{DP8 (4×4)}}$$

$$+\ 2^{-\delta_0}\,2^4\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big(a^{im}_{H,i}+(b^{re,lo}_i-8)\big)^2-\sum_{i=0}^{7}(a^{im}_{H,i}-8)^2-\sum_{i=0}^{7}(b^{re,lo}_i-8)^2+512\Big)}_{\text{DP8 (4×4)}}$$

$$+\ 2^{-\delta_0}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=0}^{7}\big((a^{im}_{L,i}-8)+(b^{re,lo}_i-8)\big)^2-\sum_{i=0}^{7}(a^{im}_{L,i}-16)^2-\sum_{i=0}^{7}(b^{re,lo}_i-16)^2+2048\Big)}_{\text{DP8 (4×4)}}$$

$$+\ 2^{-\delta_1}\,2^4\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}\big(a^{im}_{H,i}+(b^{re,lo}_i-8)\big)^2-\sum_{i=8}^{15}(a^{im}_{H,i}-8)^2-\sum_{i=8}^{15}(b^{re,lo}_i-8)^2+512\Big)}_{\text{DP8 (4×4)}}$$

$$+\ 2^{-\delta_1}\,\underbrace{\tfrac{1}{2}\Big(\sum_{i=8}^{15}\big((a^{im}_{L,i}-8)+(b^{re,lo}_i-8)\big)^2-\sum_{i=8}^{15}(a^{im}_{L,i}-16)^2-\sum_{i=8}^{15}(b^{re,lo}_i-16)^2+2048\Big)}_{\text{DP8 (4×4)}}$$

Step 9 — All components

Re (M^{re}):

$$\begin{aligned}
\text{PE} = {} & 2^{-\delta_0}\,2^8\sum_{i=0}^{7}(a^{re}_{H,i}+b^{re,hi}_i)^2+2^{-\delta_0}\,2^4\sum_{i=0}^{7}\big((a^{re}_{L,i}-8)+b^{re,hi}_i\big)^2+2^{-\delta_1}\,2^8\sum_{i=8}^{15}(a^{re}_{H,i}+b^{re,hi}_i)^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}\big((a^{re}_{L,i}-8)+b^{re,hi}_i\big)^2\\
&+2^{-\delta_0}\,2^4\sum_{i=0}^{7}\big(a^{re}_{H,i}+(b^{re,lo}_i-8)\big)^2+2^{-\delta_0}\sum_{i=0}^{7}\big((a^{re}_{L,i}-8)+(b^{re,lo}_i-8)\big)^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}\big(a^{re}_{H,i}+(b^{re,lo}_i-8)\big)^2+2^{-\delta_1}\sum_{i=8}^{15}\big((a^{re}_{L,i}-8)+(b^{re,lo}_i-8)\big)^2\\
&-2^{-\delta_0}\,2^8\sum_{i=0}^{7}(a^{im}_{H,i}+b^{im,hi}_i)^2-2^{-\delta_0}\,2^4\sum_{i=0}^{7}\big((a^{im}_{L,i}-8)+b^{im,hi}_i\big)^2-2^{-\delta_1}\,2^8\sum_{i=8}^{15}(a^{im}_{H,i}+b^{im,hi}_i)^2-2^{-\delta_1}\,2^4\sum_{i=8}^{15}\big((a^{im}_{L,i}-8)+b^{im,hi}_i\big)^2\\
&-2^{-\delta_0}\,2^4\sum_{i=0}^{7}\big(a^{im}_{H,i}+(b^{im,lo}_i-8)\big)^2-2^{-\delta_0}\sum_{i=0}^{7}\big((a^{im}_{L,i}-8)+(b^{im,lo}_i-8)\big)^2-2^{-\delta_1}\,2^4\sum_{i=8}^{15}\big(a^{im}_{H,i}+(b^{im,lo}_i-8)\big)^2-2^{-\delta_1}\sum_{i=8}^{15}\big((a^{im}_{L,i}-8)+(b^{im,lo}_i-8)\big)^2\\[1em]
\alpha = {} & 2^{-\delta_0}\,2^8\sum_{i=0}^{7}(a^{re}_{H,i})^2+2^{-\delta_0}\,2^4\sum_{i=0}^{7}(a^{re}_{L,i}-8)^2+2^{-\delta_1}\,2^8\sum_{i=8}^{15}(a^{re}_{H,i})^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}(a^{re}_{L,i}-8)^2\\
&+2^{-\delta_0}\,2^4\sum_{i=0}^{7}(a^{re}_{H,i}-8)^2+2^{-\delta_0}\sum_{i=0}^{7}(a^{re}_{L,i}-16)^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}(a^{re}_{H,i}-8)^2+2^{-\delta_1}\sum_{i=8}^{15}(a^{re}_{L,i}-16)^2\\
&-2^{-\delta_0}\,2^8\sum_{i=0}^{7}(a^{im}_{H,i})^2-2^{-\delta_0}\,2^4\sum_{i=0}^{7}(a^{im}_{L,i}-8)^2-2^{-\delta_1}\,2^8\sum_{i=8}^{15}(a^{im}_{H,i})^2-2^{-\delta_1}\,2^4\sum_{i=8}^{15}(a^{im}_{L,i}-8)^2\\
&-2^{-\delta_0}\,2^4\sum_{i=0}^{7}(a^{im}_{H,i}-8)^2-2^{-\delta_0}\sum_{i=0}^{7}(a^{im}_{L,i}-16)^2-2^{-\delta_1}\,2^4\sum_{i=8}^{15}(a^{im}_{H,i}-8)^2-2^{-\delta_1}\sum_{i=8}^{15}(a^{im}_{L,i}-16)^2\\[1em]
\beta = {} & 2^{-\delta_0}\,2^8\sum_{i=0}^{7}(b^{re,hi}_i)^2+2^{-\delta_0}\,2^4\sum_{i=0}^{7}(b^{re,hi}_i-8)^2+2^{-\delta_1}\,2^8\sum_{i=8}^{15}(b^{re,hi}_i)^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}(b^{re,hi}_i-8)^2\\
&+2^{-\delta_0}\,2^4\sum_{i=0}^{7}(b^{re,lo}_i-8)^2+2^{-\delta_0}\sum_{i=0}^{7}(b^{re,lo}_i-16)^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}(b^{re,lo}_i-8)^2+2^{-\delta_1}\sum_{i=8}^{15}(b^{re,lo}_i-16)^2\\
&-2^{-\delta_0}\,2^8\sum_{i=0}^{7}(b^{im,hi}_i)^2-2^{-\delta_0}\,2^4\sum_{i=0}^{7}(b^{im,hi}_i-8)^2-2^{-\delta_1}\,2^8\sum_{i=8}^{15}(b^{im,hi}_i)^2-2^{-\delta_1}\,2^4\sum_{i=8}^{15}(b^{im,hi}_i-8)^2\\
&-2^{-\delta_0}\,2^4\sum_{i=0}^{7}(b^{im,lo}_i-8)^2-2^{-\delta_0}\sum_{i=0}^{7}(b^{im,lo}_i-16)^2-2^{-\delta_1}\,2^4\sum_{i=8}^{15}(b^{im,lo}_i-8)^2-2^{-\delta_1}\sum_{i=8}^{15}(b^{im,lo}_i-16)^2\\[1em]
C = {} & \big(2^{-\delta_0}+2^{-\delta_1}\big)\cdot 18432-\big(2^{-\delta_0}+2^{-\delta_1}\big)\cdot 18432=0\\
M^{re} = {} & \tfrac{1}{2}\big(\text{PE}-\alpha-\beta+C\big),\qquad E=\max(E_0,E_1)
\end{aligned}$$

Im (M^{im}):

$$
\begin{aligned}
\mathrm{PE} = {} & 2^{-\delta_0}\,2^8\sum_{i=0}^{7}(a^{re}_{H,i}+b_i^{im,hi})^2+2^{-\delta_0}\,2^4\sum_{i=0}^{7}\big((a^{re}_{L,i}-8)+b_i^{im,hi}\big)^2+2^{-\delta_1}\,2^8\sum_{i=8}^{15}(a^{re}_{H,i}+b_i^{im,hi})^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}\big((a^{re}_{L,i}-8)+b_i^{im,hi}\big)^2 \\
& +2^{-\delta_0}\,2^4\sum_{i=0}^{7}\big(a^{re}_{H,i}+(b_i^{im,lo}-8)\big)^2+2^{-\delta_0}\sum_{i=0}^{7}\big((a^{re}_{L,i}-8)+(b_i^{im,lo}-8)\big)^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}\big(a^{re}_{H,i}+(b_i^{im,lo}-8)\big)^2+2^{-\delta_1}\sum_{i=8}^{15}\big((a^{re}_{L,i}-8)+(b_i^{im,lo}-8)\big)^2 \\
& +2^{-\delta_0}\,2^8\sum_{i=0}^{7}(a^{im}_{H,i}+b_i^{re,hi})^2+2^{-\delta_0}\,2^4\sum_{i=0}^{7}\big((a^{im}_{L,i}-8)+b_i^{re,hi}\big)^2+2^{-\delta_1}\,2^8\sum_{i=8}^{15}(a^{im}_{H,i}+b_i^{re,hi})^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}\big((a^{im}_{L,i}-8)+b_i^{re,hi}\big)^2 \\
& +2^{-\delta_0}\,2^4\sum_{i=0}^{7}\big(a^{im}_{H,i}+(b_i^{re,lo}-8)\big)^2+2^{-\delta_0}\sum_{i=0}^{7}\big((a^{im}_{L,i}-8)+(b_i^{re,lo}-8)\big)^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}\big(a^{im}_{H,i}+(b_i^{re,lo}-8)\big)^2+2^{-\delta_1}\sum_{i=8}^{15}\big((a^{im}_{L,i}-8)+(b_i^{re,lo}-8)\big)^2 \\[6pt]
\alpha = {} & 2^{-\delta_0}\,2^8\sum_{i=0}^{7}(a^{re}_{H,i})^2+2^{-\delta_0}\,2^4\sum_{i=0}^{7}(a^{re}_{L,i}-8)^2+2^{-\delta_1}\,2^8\sum_{i=8}^{15}(a^{re}_{H,i})^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}(a^{re}_{L,i}-8)^2 \\
& +2^{-\delta_0}\,2^4\sum_{i=0}^{7}(a^{re}_{H,i}-8)^2+2^{-\delta_0}\sum_{i=0}^{7}(a^{re}_{L,i}-16)^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}(a^{re}_{H,i}-8)^2+2^{-\delta_1}\sum_{i=8}^{15}(a^{re}_{L,i}-16)^2 \\
& +2^{-\delta_0}\,2^8\sum_{i=0}^{7}(a^{im}_{H,i})^2+2^{-\delta_0}\,2^4\sum_{i=0}^{7}(a^{im}_{L,i}-8)^2+2^{-\delta_1}\,2^8\sum_{i=8}^{15}(a^{im}_{H,i})^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}(a^{im}_{L,i}-8)^2 \\
& +2^{-\delta_0}\,2^4\sum_{i=0}^{7}(a^{im}_{H,i}-8)^2+2^{-\delta_0}\sum_{i=0}^{7}(a^{im}_{L,i}-16)^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}(a^{im}_{H,i}-8)^2+2^{-\delta_1}\sum_{i=8}^{15}(a^{im}_{L,i}-16)^2 \\[6pt]
\beta = {} & 2^{-\delta_0}\,2^8\sum_{i=0}^{7}(b_i^{im,hi})^2+2^{-\delta_0}\,2^4\sum_{i=0}^{7}(b_i^{im,hi}-8)^2+2^{-\delta_1}\,2^8\sum_{i=8}^{15}(b_i^{im,hi})^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}(b_i^{im,hi}-8)^2 \\
& +2^{-\delta_0}\,2^4\sum_{i=0}^{7}(b_i^{im,lo}-8)^2+2^{-\delta_0}\sum_{i=0}^{7}(b_i^{im,lo}-16)^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}(b_i^{im,lo}-8)^2+2^{-\delta_1}\sum_{i=8}^{15}(b_i^{im,lo}-16)^2 \\
& +2^{-\delta_0}\,2^8\sum_{i=0}^{7}(b_i^{re,hi})^2+2^{-\delta_0}\,2^4\sum_{i=0}^{7}(b_i^{re,hi}-8)^2+2^{-\delta_1}\,2^8\sum_{i=8}^{15}(b_i^{re,hi})^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}(b_i^{re,hi}-8)^2 \\
& +2^{-\delta_0}\,2^4\sum_{i=0}^{7}(b_i^{re,lo}-8)^2+2^{-\delta_0}\sum_{i=0}^{7}(b_i^{re,lo}-16)^2+2^{-\delta_1}\,2^4\sum_{i=8}^{15}(b_i^{re,lo}-8)^2+2^{-\delta_1}\sum_{i=8}^{15}(b_i^{re,lo}-16)^2 \\[6pt]
C = {} & \big(2^{-\delta_0}+2^{-\delta_1}\big)\cdot 18432+\big(2^{-\delta_0}+2^{-\delta_1}\big)\cdot 18432 = \big(2^{-\delta_0}+2^{-\delta_1}\big)\cdot 36864 \\
M^{im} = {} & \tfrac{1}{2}\big(\mathrm{PE}-\alpha-\beta+C\big), \qquad E = \max(E_0, E_1)
\end{aligned}
$$